

Part A · Multiple Choice

2 points each

1. **a) Its focus on spatial inquiry and analysis**

p. 9 — “What makes geography different from other disciplines is its focus on spatial inquiry and analysis.”

2. **d) dependent variable**

p. 13 — erosion is what is being measured, so it depends on the farming technique (the independent variable). Slope, water, and fertilizer are the controls.

3. **b) it explains all the collected data and none of the data contradicts it**

p. 13 — choice (a) is deliberately wrong: the text states that science does not prove anything beyond a shadow of a doubt.

4. **c) The Equator**

pp. 14–15 — the Equator is the only parallel that is also a great circle. All other parallels are small circles.

5. **a) an angular measurement north or south of the equator**

p. 15 — the text explicitly corrects the common assumption that latitude is a line. The line connecting points of equal latitude is a parallel.

6. **b) 15 degrees of longitude**

p. 17 — $360^\circ \div 24 \text{ hours} = 15^\circ \text{ per hour}$, or 7.5° on each side of the zone’s center line.

Part B · True or False

1 point each

7. **FALSE**

Longitude is measured east and west of the Prime Meridian at Greenwich. The International Date Line lies at 180° from it. (p. 16)

8. **TRUE**

p. 18 — human eyes are given as an example of passive remote sensing.

Part C · Short Answer

3 points each

9. **Absolute location is an exact coordinate position on Earth — a specific x- and y- value. Relative location describes where something is in relation to other things.**

Rubric: 1 pt correct definition of absolute; 1 pt correct definition of relative; 1 pt for a valid example of each. Accept any reasonable examples — the book uses coordinates typed into Google Maps (absolute) versus a driving instruction such as “turn left at the Valencia Blvd exit,” or giving directions to your house using landmarks (relative). (p. 14)

10. **Raster and vector.**

Rubric: 1 pt for naming raster; 1 pt for naming vector; 1 pt for a correct distinction. Raster models are grid images made of rows and columns of pixels, each georeferenced and holding a numeric value — most remote sensing imagery enters a GIS this way. Vector models are built from georeferenced points, which connect into lines and close into areas (polygons), each carrying an attached database. Full credit for capturing “pixels/grid” versus “points, lines, and polygons.” (p. 21)